

AYBERK YARKIN YILDIZ

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Education

Northeastern University, Boston, MA, USA

Fall 2023 – Spring 2028

Doctor of Philosophy in Computer Engineering

CGPA: 3.84/4.00

- Courses: Verifiable Machine Learning, Advanced Computer Vision

Northeastern University, Boston, MA, USA

Fall 2023 – Spring 2025

Master of Science in Electrical and Computer Engineering

CGPA: 3.84/4.00

- Courses: Advanced Machine Learning, Probabilistic System Modeling, Machine Learning with Small Data, Introduction to Machine Learning and Pattern Recognition, Data Visualization, Parallel Processing for Data Analytics, Applied Probability and Stochastic Processes, Fundamentals of Computer Engineering

Bilkent University, Ankara, Turkey

Fall 2018 – Spring 2023

Bachelor of Science in Electrical and Electronics Engineering

Friedrich-Alexander-Universität, Erlangen, Germany

Spring 2022

Erasmus Student in Elektrotechnik – Elektronik und Informationstechnik

Mehmet Emin Resulzade High School, Ankara, Turkey

Fall 2014 – Spring 2018

- *CGPA: 98/100.* Attended Nesibe Aydın High School for the last year.

Skills

Programming: Python (PyTorch, Tensorflow, PySpark), MATLAB, C/C++

Languages: English (Fluent – C1 level), Turkish (Native Fluency), German (Beginner – A2 level)

Research and Work Experience

Northeastern University

Boston, MA, USA

Graduate Research Assistant

Fall 2023 – Spring 2028

- Research on scalable, communication-aware optimization for pruning, mapping, and compressing deep neural networks aligned with hardware topologies to enable resilient, energy-efficient, and low-latency distributed inference, as well as optimal data selection strategies via Markovian experimental design for re-training machine learning models to maintain predictive performance in the presence of concept drifts.

Neurocess Limited

London, England

Data Science / Machine Learning Engineer (Remote)

Fall 2022 – Summer 2023

- Analyzed data from active **sEMG** sensors for monitoring athletic and physiologic performance of football players using **Siamese networks** with **CNNs** and **Transformers** for multivariate time-series implemented in **PyTorch** and **Keras**.

KOCLAB, National Magnetic Resonance Research Center (UMRAM)

Ankara, Turkey

Undergraduate Researcher

Fall 2021 – Fall 2022

- Conducted deep learning research on implementation and technical extensions of time-series analysis and imputation using **RNNs** and **Transformer** models in **PyTorch**.

Bilkent University

Ankara, Turkey

Teaching Assistant

Spring 2020 – Fall 2021

- Tutored students and graded assignments for Calculus I, Calculus II for 60 students per semester.

UMRAM

Ankara, Turkey

Intern

Summer 2020

- Implemented and tested the interfaces of the fundamental electronic devices such as a gaussmeter, an analog filter, and a DC power supply in **MATLAB** for the company's future research. Led to a publication in Medical Physics. 📄

Projects

Multi-Agent Distributed Inference over Large AI Models at the Network Edge

Fall 2025

- Developed a **compression-driven Model-Distributed Inference (MDI)** framework for executing **large AI models** across heterogeneous **edge devices**, leveraging lightweight **pruning** and **quantization** to reduce communication overhead under **wireless constraints**, validated on **Raspberry Pi** and **NVIDIA Jetson Nano** testbeds.

Communication-aware neural Mapping and Pruning Framework

Spring 2024 – Spring 2025

- Performed efficient distributed inference over communication-aware mapped and pruned **CNNs** using **PyTorch**. Tested on real-life environments such as **Colosseum** wireless emulator in wired, wireless, and cellular scenarios, and **Raspberry Pi** testbeds. Observed up to a **26×** speedup over pruned models.

- Markovian Experimental Design under Concept Drift
Spring 2024 – Spring 2025

 - Implemented optimal data selection strategies for re-training ML models to maintain predictive performance in the presence of concept drifts via **Kalman filters** in **Python**.
- Gradient Boosting Decision Trees on Medical Diagnosis
View publication [4]
Fall 2024

 - Implemented an extensive analysis of **ensemble models** in medical diagnosis focusing on the superior performance over state-of-the-art deep learning models in **PyTorch**.
- Wireless Radar Classification with Transformers
View preprint [5]
Fall 2023 – Spring 2024

 - Implemented **Transformer**-based classification models for wireless radar signals over out-of-distribution data with **LoRA** and **conformal prediction** in **PyTorch**.
- Portable RF Signal Sensing System Using SDR
View publication [6]
Fall 2023

 - Implemented an Electronic Support Measures (ESM) system with a **GPU** accelerated **SDR** that could detect, measure, and classify **RF** signals using signal processing algorithms in **GNU Radio** and **XGBoost**.
- sEMG Motion Classification and Anomaly Detection
View preprint
Fall 2022 – Fall 2023

 - Implemented motion classification and anomaly detection models using **sEMG** signals via few-shot learning with a **Siamese network** in **Keras** and **PyTorch**.
- Multivariate Time Series Imputation With Transformers
View publication [7]
Fall 2022

 - Developed a **Transformer**-based **autoencoder** for missing value imputation that outperformed seven state-of-the-art imputation methods by **13.5 – 50.5%** over benchmark datasets.

Publications

[1]

A. Y. Yıldız*, M. Sirera Perelló*, T. Li*, C. Li, P. Gholami, S. Reddy, T. T. Davarakis, A. Arora, K. Chowdhury, A. Eryılmaz, S. Ioannidis, H. Seferoglu, and N. Shroff, “Multi-Agent Distributed Inference over Large AI Models at the Network Edge.” (submitted to IEEE Wireless Communications Magazine)

[2]

A. Y. Yıldız*, M. Sirera Perelló*, T. T. Davarakis*, J. Groen, H. Seferoglu, K. Chowdury, and S. Ioannidis, “CaMP: Communication-Aware Neural Mapping and Pruning.” (submitted to IEEE INFOCOM 2026)

[3]

A. Y. Yıldız, L. Su, C. Joe-Wong, E. Yeh, and S. Ioannidis, “Markovian Experimental Design under Concept Drift.” (submitted to AISTATS 2026)

[4]

A. Y. Yıldız and A. Kalayci, “Gradient boosting decision trees on medical diagnosis over tabular data.” 2025 IEEE International Conference on AI and Data Analytics (ICAD). IEEE, 2025.

[5]

M. Belgiovine, J. Groen, M. Sirera, C. Tassie, **A. Y. Yıldız**, S. Ioannidis, and K. Chowdhury, “T-PRIME: Real-time Deployment of a Transformer-based Protocol Identification for Machine-learning at the Edge.” arXiv preprint arXiv:2401.04837 (2024). (submitted to IEEE/ACM Transactions on Networking)

[6]

G. S. Yavuz, B. Saygılı, Y. Aydın, R. Dalkıran, İ. Eşin, M. Uluçay, B. Uykulu, S. S. Kıyma, O. Arikan, and **A. Y. Yıldız**, “Detection and classification architecture for sdr based radar electronic support measure systems.” 2024 32nd Signal Processing and Communications Applications Conference (SIU). IEEE, 2024.

[7]

A. Y. Yıldız, E. Koç, and A. Koç. “Multivariate time series imputation with transformers.” IEEE Signal Processing Letters 29 (2022): 2517-2521.

Honors & Awards

- Invited Talks
 - IEEE Signal Processing Society Blog2024
 - IEEE International Conference on Image Processing2023
 - IEEE Signal Processing Society’s top 25 downloaded articlesSep 2022 – Sep 2023
 - Research Excellence Award at Bilkent University2023
 - High Honor Student & Tuition Scholarship at Bilkent University2018 – 2023

Extracurricular Activities & Memberships

- Amazon IET-MSI Program ParticipantFall 2024
 - Northeastern University SPIRAL Committee MemberFall 2024
 - Organized seminars with participants from universities in USA and Switzerland.
 - Bilkent University Community Awareness Projects Active MemberSpring 2021
 - 8/8th grade practical and theoretical certificate for piano, ABRSM, London, UKFebruary 2023